

# MANIPULATING/TRANSFORMING DATA IN SPSS

STC224

# Introduction (SPSS windows)

- **The Start Screen**
- **The Data Editor Window**  
Where you create the template (enter variables) and enter the data
- **The Output Viewer Window**  
Where you view and edit output produced by a data analysis command e.g. charts, tables etc.
- **The Syntax Editor Window**  
Where you enter and run custom command syntax (SPSS language)
- **The Script Editor Window**  
Where you type scripts (in BASIC and Python) that extend the functionality of IBM SPSS Statistics e.g. automating some features like opening and saving data files

# Introduction cont'd

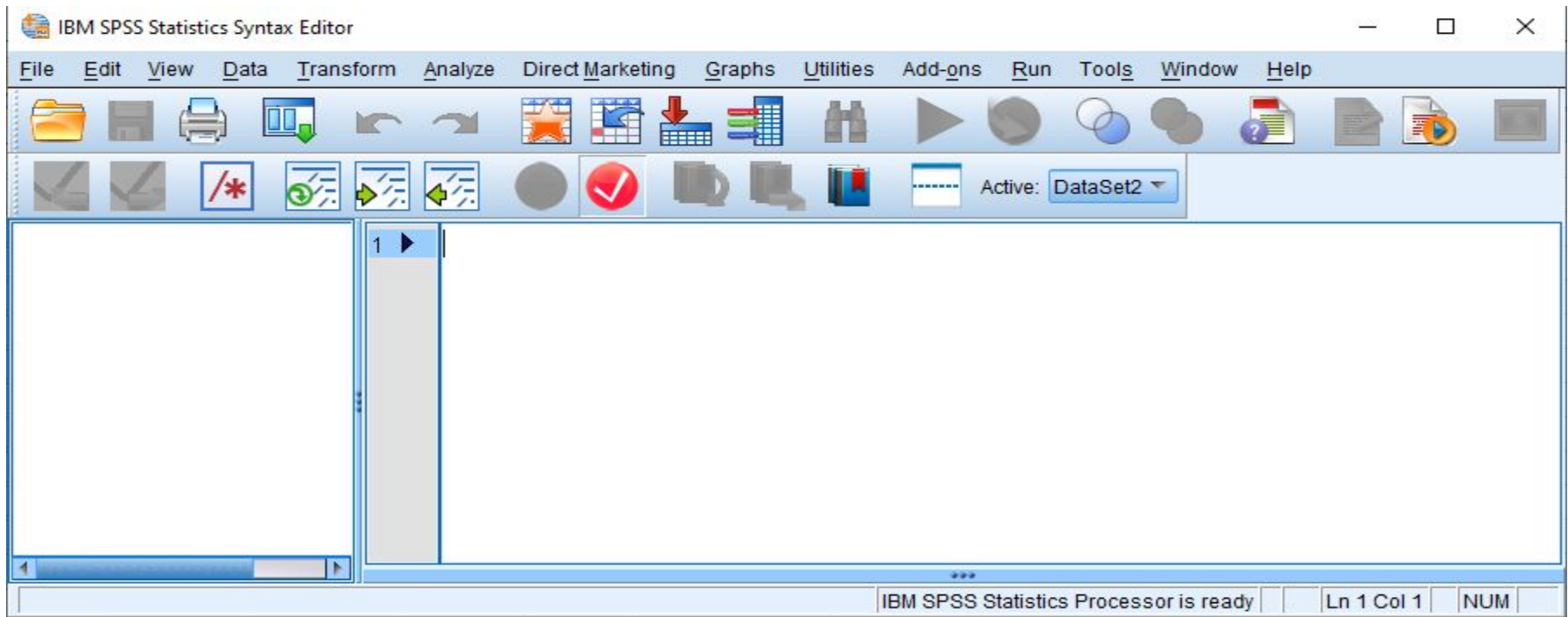
- Measure – The levels of measurement of the variable
  - **Nominal** : There is no intrinsic ranking to the categories e.g. Sex, religious affiliation
  - **Ordinal** : There is intrinsic order to the categories e.g. level of satisfaction (satisfied, highly satisfied) or confidence (agree, strongly agree)
  - **Scale** : Continuous values with meaningful metric e.g. salary, age

# Creating files in the Syntax Editor Window

- An alternate way to create data files is to use the **SPSS Syntax Editor Window**.
- The Syntax Editor works much like a word processor, such as Windows Notepad or a scaled-down version of Microsoft Word.
- That is, you can just type the data into this window as you would in creating any text file.

# Accessing the Syntax Editor

- Select **File, New, Syntax** from the drop-down menu at the top of the Data Editor window as shown below
- A new window will open



# Syntax Editor cont'd

- Note that in addition to typing text/data directly into the Syntax Editor window, you can "cut-and-paste" data into this editor.
- That is, just as you can cut-and-paste text from one Word file into another Word file, you can also cut-and-paste data into the Syntax Editor.
- Thus, data files that were created in a text editor such as Windows Notepad (called ASCII files) can be cut-and-pasted into this Syntax Editor.
- However, as mentioned, we will create this Syntax Editor file from scratch by typing the text and data directly into this editor.

# Entering Commands & Data in the Syntax File

- We will be working with the Data Editor window, so we will not discuss the Syntax Editor in detail here.
- For now, just type the file exactly as it appears in in Figure below into your blank Syntax Editor.

# Entering Commands & Data in the Syntax File cont'd

SyntaxWin1.sps - IBM SPSS Statistics Syntax Editor

The screenshot shows the IBM SPSS Statistics Syntax Editor window. The menu bar includes File, Edit, View, Data, Transform, Analyze, Direct Marketing, Graphs, Utilities, Add-ons, Run, Tools, Window, and Help. The toolbar contains various icons for file operations, editing, and running syntax. The main window displays a syntax file with the following content:

```
data list
begin data.
frequencies
execute.

1 data list records=1/1 id 1-2 sex 3 courses 4.
2 begin data.
3 0112
4 0211
5 0311
6 0412
7 0513
8 0623
9 0723
10 0822
11 0924
12 1023
13 end data.
14 frequencies variables=all.
15 execute.
16
```

A yellow callout box points to the first three lines of the syntax file, containing the text: "Type these lines into your syntax window. Don't forget to add periods at the ends of the first two and last three lines!"



# Entering Commands & Data in the Syntax File cont'd

data list records=1/1 id 1-2 sex 3 courses 4.

begin data.

0112

0211

0311

0412

0513

0623

0723

0822

0924

1023

end data.

frequencies variables=all.

execute.

# Entering Commands & Data in the Syntax File cont'd

## Explanation:

- **data list** — tells SPSS that you are about to *manually enter data*.
- **records=1/1** : means that **each case (row)** has **1 record (line)** of data.
- **id 1-2** : variable **id** occupies **columns 1 and 2** (two digits).
- **sex 3** : variable **sex** occupies **column 3** (one digit).
- **courses 4** : variable **courses** occupies **column 4** (one digit).

# Entering Commands & Data in the Syntax File cont'd

- The first line of this file consists of syntax code, or SPSS commands, called the **DATA LIST** statement.
- This instructs SPSS how to read the data in the file.
- Note that the three variables (**ID, Sex, and Courses**) are defined on this line.
- The actual lines of data (scores) are always sandwiched between a **Begin Data** and an **End Data** syntax commands.
- That is, each line of data (or row) consists of each student's ID number, code for sex, and number of psychology courses.

# Entering Commands & Data in the Syntax File cont'd

- The last two lines instruct SPSS to generate frequency tables of each variable.
- Note that except for the lines listing the data, the lines containing syntax code must all end with a period.
- After you have created your file, you will be ready to save it as a syntax file.
- **Saving Files in the Syntax Editor: The .SPS File extension is used**

# Generating an output file

- Let's go ahead and run the Frequencies procedure specified in this Syntax file so that we can generate an output file to view.
- To generate this output, we need to instruct SPSS to run the procedure specified in the Frequencies syntax line at the end of the file.
- To do this, select **Run, All** from the dropdown menu at the top of this Syntax Editor Window

# MANIPULATING/TRANSFORMING DATA

- Recoding
- Binning
- Computing
- Selecting (Excluding)
- Splitting

# Recoding (Transforming) Variables

- Sometimes you will want to transform a variable by combining some of its categories or values together.
- For example, you may want to change a continuous variable into an ordinal categorical variable, or you may want to merge the categories of a nominal variable.
- In SPSS, this type of transform is called ***recoding***.
- In SPSS, there are three basic options for recoding variables:
  - a) Recode into Different Variables
  - b) Recode into Same Variables
  - c) DO IF syntax

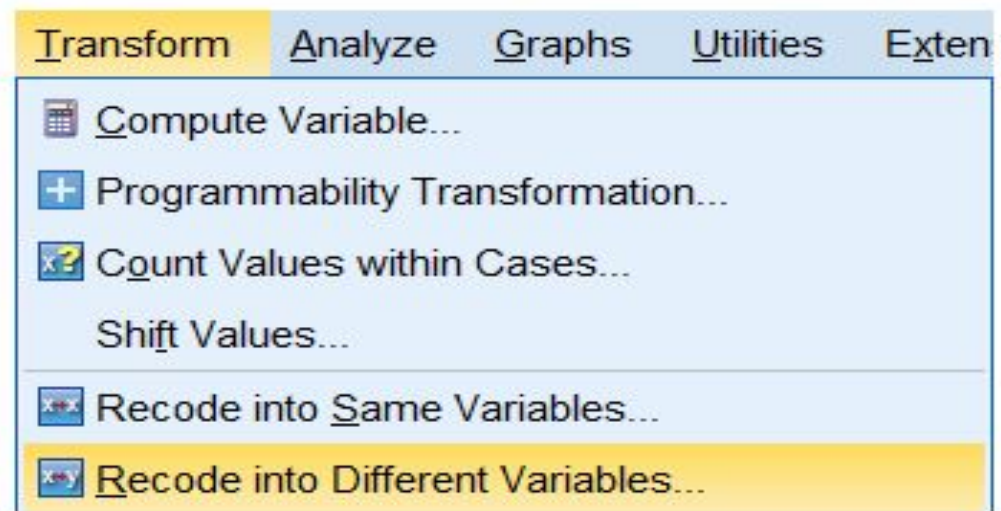
# Recoding (Transforming) Variables cont'd

- Each of these options allows you to re-categorize an existing variable.
- **Recode into Different Variables** and **DO IF** syntax create a new variable without modifying the original variable, while **Recode into Same Variables** will permanently overwrite the original variable.
- In general, it is best to recode a variable into a ***different*** variable so that you never alter the original data and can easily access the original data if you need to make different changes later on



# Recode into Different Variables

- Recoding into a different variable transforms an original variable into a new variable.
- That is, the changes do not overwrite the original variable; they are instead applied to a copy of the original variable under a new name.



# Example

- Class ranks for high schools and colleges are nicknames for what year of study the person is completing: "**freshman**" (first-year), "**sophomore**" (second-year), "**junior**" (third-year), "**senior**" (fourth-year).
- Class ranks are also sometimes divided into "**underclassmen**" (first or second-year students) and "**upperclassmen**" (third or fourth-year students).
- In the sample dataset, the variable Rank has the categories Freshman (1), Sophomore (2), Junior (3), and Senior (4).
- Let's use Recode into Different Variables to merge the categories and create a new indicator variable called ***RankIndicator*** with the levels Underclassman (1) and Upperclassman (2).

# Solution

- Using the Dialog Windows
- This method tells SPSS exactly how to map each old category onto a new category.
- Click **Transform > Recode into Different Variables**.
- Double-click on variable Rank to move it to the Input Variable -> Output Variable box. In the Output Variable area, give the new variable the name ***RankIndicator***. Define the label as *Class Rank (binary)*, and then click **Change**.
- Click the **Old and New Values** button.
  - Handle missing values first: In the Old Value area click **System-missing**; in the New Value area click **System-missing**. Then click **Add**.
  - Define the underclassmen group (1):
    - In the Old Value area click **Value** and enter **1**; in the New Value area click **Value** and enter **1**. Then click **Add**.
    - In the Old Value area click **Value** and enter **2**; in the New Value area click **Value** and enter **1**. Then click **Add**.
  - Define the upperclassmen group (2):
    - In the Old Value area click **Value** and enter **3**; in the New Value area click **Value** and enter **2**. Then click **Add**.
    - In the Old Value area click **Value** and enter **4**; in the New Value area click **Value** and enter **2**. Then click **Add**.
  - When finished, click **Continue**.
- Click **OK**.

# Recode into Same Variables

- Recoding into the same variable (**Transform > Recode into Same Variables**) works the same way as described above, except for that any changes made will permanently alter the original variable.
- That is, the original values will be replaced by the recoded values.
- In general, it is good practice *not* to recode into the same variable because it overwrites the original variable. If you ever needed to use the variable in its original form (or wanted to double-check your steps), that information would be lost.

# Data editing Visual Binning

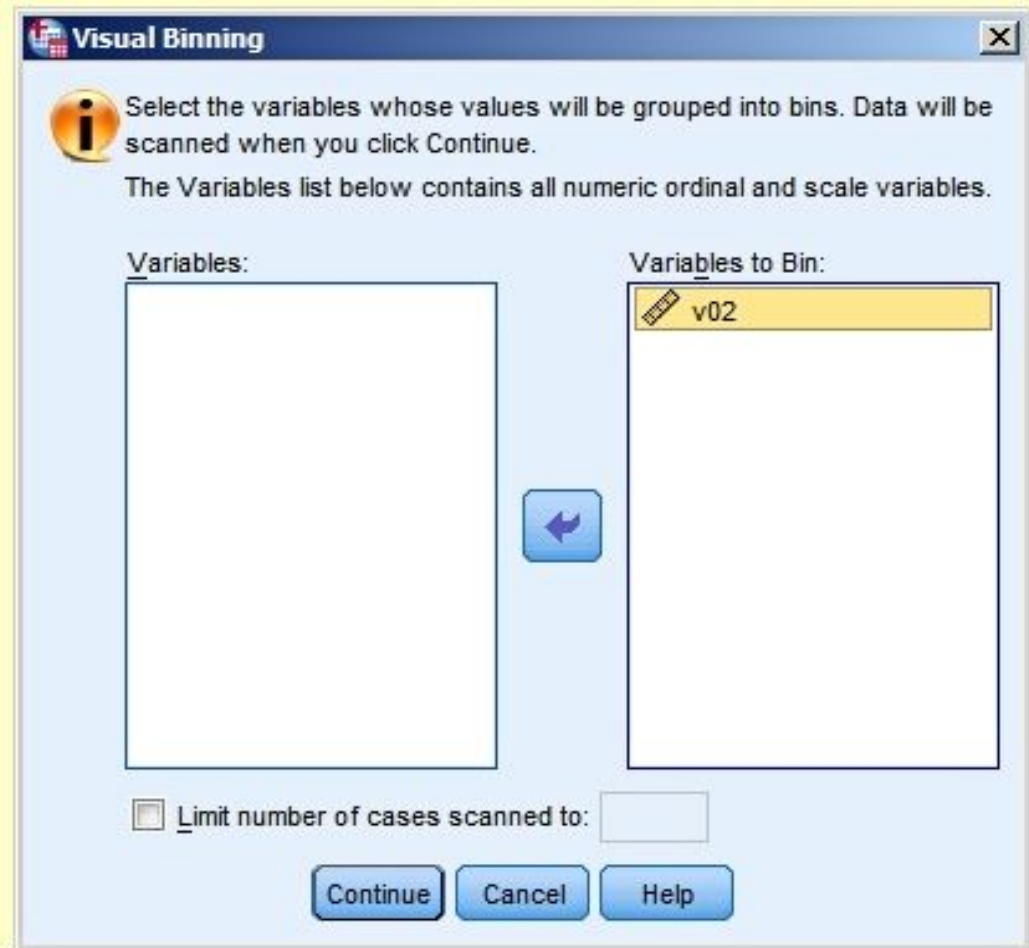
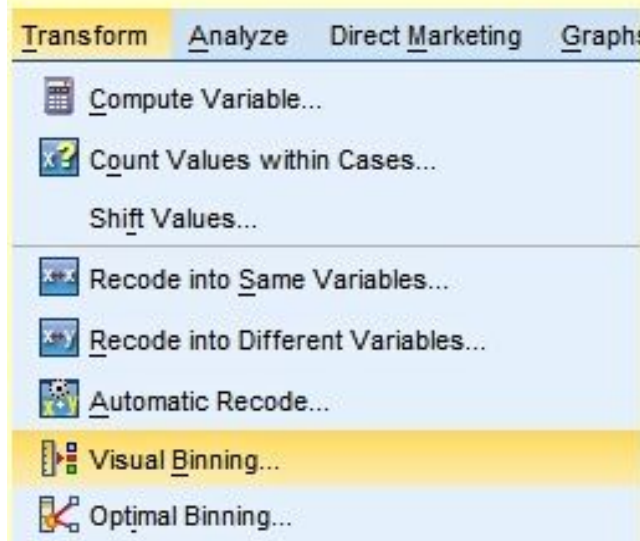
- Through Visual Binning you can in an easy and visual way create classes for a scale variable.
- The labels for the classes are automatically added by SPSS.
- In our example data file the variable v02 contains the age of the respondent in years.

## Coding

| Name | Type    | Width | Decimals | Label        | Values          | Missing | Columns | Align   | Measure                                                                                     | Role                                                                                       |
|------|---------|-------|----------|--------------|-----------------|---------|---------|---------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| v01  | Numeric | 8     | 0        | Gender       | {1, Male}...    | 9       | 8       | ≡ Right |  Nominal |  Input  |
| v02  | Numeric | 3     | 0        | Age in years | {999, Unknow... | 999     | 8       | ≡ Right |  Scale  |  Input |

# Using Visual Binning

- Choose **Transform > Visual Binning** and select variable **v02**.



# Data editing Visual Binning cont'd

- Click Continue.
- Now you come to the actual Visual Binning dialog box. Complete it as shown below, following steps 1, 2 and 3.

The image shows the SPSS Visual Binning dialog box. The 'Scanned Variable List' on the left contains 'v02'. The 'Current Variable' is 'v02' and the 'Binned Variable' is 'v02\_binned'. The 'Label' for the current variable is 'Age in years' and for the binned variable is 'Age in years (Binned)'. The 'Minimum' is 18 and the 'Maximum' is 55. A histogram shows the distribution of ages. The 'Grid' table shows the resulting bins. The 'Upper Endpoints' section has 'Included (<=)' selected. The 'Make Cutpoints...' and 'Make Labels' buttons are highlighted with red boxes and labeled 'Step 1' and 'Step 2' respectively. The 'Minimum' field is highlighted with a red box and labeled 'Step 3'.

**Visual Binning**

Scanned Variable List: v02

Name: Current Variable: v02 Label: Age in years

Binned Variable: v02\_binned Label: Age in years (Binned)

Minimum: 18 Nonmissing Values Maximum: 55

**Step 3**

Enter interval cutpoints or click Make Cutpoints for automatic intervals. A cutpoint value of 10, for example, defines an interval starting above the previous interval and ending at 10.

Grid:

|   | Value | Label   |
|---|-------|---------|
| 1 | 21.0  | <= 21   |
| 2 | 29.5  | 22 - 30 |
| 3 | 38.0  | 31 - 38 |
| 4 | 46.5  | 39 - 47 |
| 5 | HIGH  | 48+     |
| 6 |       |         |

Upper Endpoints  
☒ Included (<=)  
☐ Excluded (<)

**Step 1** Make Cutpoints...

**Step 2** Make Labels

☐ Reverse scale

OK Paste Reset Cancel Help

# Compute

- Sometimes you might want to create a new variable based on some mathematical calculations of two or more variables. E.g. adding all household members to a household members total.
- To compute a new variable by summing scores on several variables, simply select **Transform, Compute** from the Data Editor menu



# File Splitting

- The split file procedure is used when we want to run the analysis on several groups or segments of the entire population. (The Split File command is used to separate the output of SPSS tests according to a group variable.)
- The best example would be to split the output according to sex.
- By doing this in SPSS, through the use of the Split File command, you will get two separate outputs for subsequent analyses, one for males and the other for females.
- To be able to perform a split file command, you must have a grouping (nominal) variable, such as sex (male/female), age (young/old), treatment group (control/treated) etc.
- There can be more than two groups within the variable.

# How to use the Split File command in SPSS

- To use the Split File command within SPSS, firstly go to **Data > Split File ....**
- A new window should open. Here, select the **Organize output by groups** option. Now you can drag the grouping variable you want to split the file by into the box called **Groups Based on:.**
- Click the **OK** button. Any analyses performed from now on will be split into separate outputs based on the grouping variable.

# Selecting cases (excluding)

- For some reasons, you might want to select some cases and exclude others from your analyses e.g.
  - They are extreme outliers
  - You only want to produce charts or some other analyses that contain a count of cases that gave a particular response to a question e.g. only those who said they have a bicycle or only those that are below 45 years of age.

# Selecting cases (excluding) cont'd

- To select only those 45 years old and below
- Go to **“Data”, “Select Cases”**
- On the **“Select”** box, choose **“If condition is satisfied”**. Click **[If]**.
- From the variable list, select **“age”** and move it to the selection box.
- On the virtual Keypad, click **[<=]** (Less or equal to). Then type **45**. Then click **[Ok]**
- Cases with age 45 and above will be crossed out and will not be included in any analysis

# Selecting cases (excluding) cont'd

- To select only those who said they have a bicycle
- First clear out the last filter by going to “Data”, “Select Cases”, and choose “All cases” and click [Ok]
- Go to “Data”, “Select Cases”
- On the “Select” box, choose “If condition is satisfied”. Click [If].
- From the variable list, select “has Bicycle” and move it to the selection box.
- Since this is a dichotomous variable, those who said they have a bicycle indicated 1.
- Hence on the virtual Keypad, click [=] (equals). Then type 1. Then click [Ok]
- Respondents without bicycles will be crossed out and will not be included in any analysis

# Questions

